

ECE253/CSE208 Introduction to Information Theory

Lecture 1: Introduction & Overview

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ECE Department

University of California, Santa Cruz

- Syllabus of this course
- Chap 1 of *Elements of Information Theory (2nd Edition)* by Thomas Cover & Joy Thomas.

Outline

- Course Information
- Overview of Information Theory

Course Information

Lecture Time and Location

- TuTh 3:20-4:55pm at E2-192.
- Course websites: Canvas & Gradescope.

Instructor: Yu Zhang (zhangy@ucsc.edu). Office hour: Mondays 3-4pm at E2-547B

Supporting service for students with disabilities: Please contact disability resource center (DRC, 831-459-2089, drc@ucsc.edu) for more information.

- Students contact the DRC to request and receive their accommodation letter.
- Students manage their own disclosure of disability status.

Objectives and Prerequisites

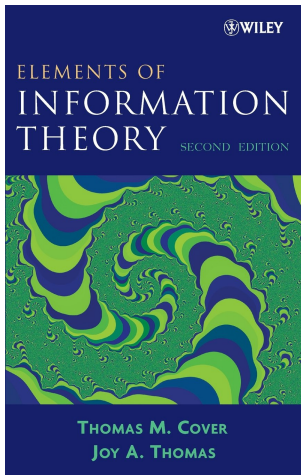
Objectives:

- To understand the concepts of entropy, mutual information and channel capacity.
- To enhance the knowledge of probability and measures of information.
- To guide the student through the implications and consequences of fundamental theories and laws of information/coding theory regarding many applications in communications and machine learning.

Prerequisites:

1. Probability and stochastic processes, multivariate calculus, and theorem-proof exposition.
2. Optional: ECE255 Error Control Coding.

Mandatory Textbook R1



- Canonical graduate textbook that established the standard modern formulation of information theory
- Rigorous and comprehensive, with precise definitions, proofs, and theorem-driven development
- Strong foundations in probability and entropy, carefully building toward coding theorems
- Broad coverage including source coding, channel capacity, rate–distortion, and network information theory

Optional Textbooks S1–S2

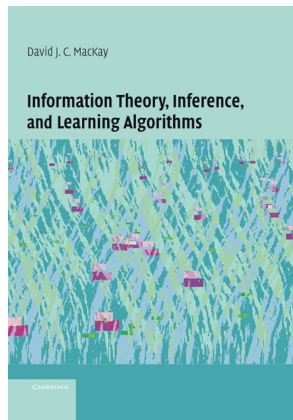
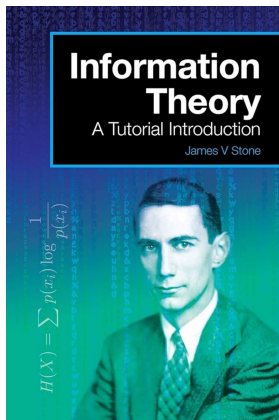


Figure: S1) Most intuitive entry point; S2) Unifies information theory with inference and learning.

Optional Textbooks S3–S4

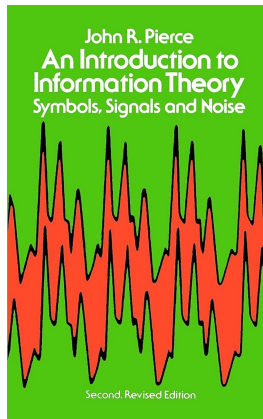
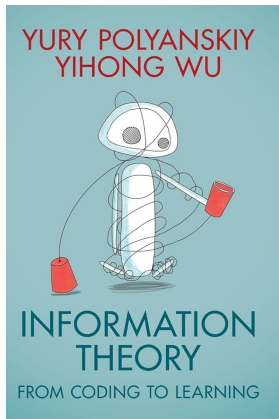


Figure: S3) Modern, rigorous, research-level treatment linking coding and learning; S4) Conceptual and historical perspective.

Tentative Agenda

Week	Topic (R1's chapters 1-10; chap 17 is integrated throughout this course)
Week-1	Course overview; probability theory revisited
Week-2,3	Entropy, mutual information, and inequalities (chap 2)
Week-4	AEP, entropy rate (chap 3-4)
Week-5,6	Data compression and gambling (chap 5-6)
Week-7	Channel capacity (chap 7);
Week-8	Differential entropy (chap 8); Gaussian channel I (chap 9)
Week-9	Gaussian channel II (chap 9)
Week-10	Rate distortion theory (chap 10)

Key dates:

- In-class Midterm Exam: Thursday, Feb 12, 2026
- Take-home (24-hour) Final Exam: Tuesday, Mar 17, 2026

Grading Policy

Workload

Homework	30%
Midterm exam	40%
Final exam	30%

Grading scale: satisfactory = B or above.

[93, 100]	<i>A</i> ⁺
[86, 92]	<i>A</i>
[80, 85]	<i>A</i> ⁻
[75, 79]	<i>B</i> ⁺
[70, 74]	<i>B</i>
[65, 69]	<i>C</i>
[60, 64]	<i>D</i>
< 60	Fail

Homework & Exam Policy

1. Assignments and submissions are through Gradescope only.
2. No late submissions are allowed.
3. Instructor-granted extensions are granted only in exceptional circumstances. Students must notify the instructor by email before the deadline, explaining the situation and the requested extension duration.

Academic integrity

- All assignments that are turned in must represent your own work (no plagiarism).
- Please ask for clarification when you are unsure if a particular behavior violates the policy.
- Complete information about academic integrity policy.

Overview of Information Theory

What Is Information Theory?

- **Information theory** is the mathematical study of *information*:
 - How information is **represented**
 - How information is **transmitted**
 - How information is **stored and compressed**
- It provides fundamental limits on:
 - **Efficiency**: How fast we can communicate
 - **Reliability**: How reliably we can communicate
 - **Compression**: How much data can be compressed without losing meaning ([Entropy H](#))
- These limits are universal:
 - Independent of hardware, algorithms, or technology
 - Apply to communication, data storage, learning, and inference

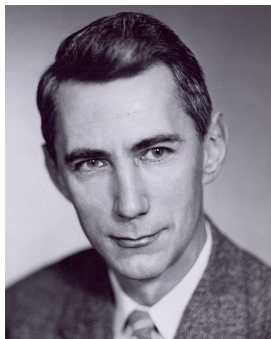
A Simple Intuition

A basic question

How much information can be sent *reliably* over a noisy channel?

- Noise corrupts signals:
 - Wireless interference
 - Measurement errors
 - Random disturbances
- Information theory shows that:
 - Perfect reliability is possible *below* a certain rate
 - Reliable communication is impossible *above* that rate
- This critical limit is called the **Channel capacity C**

Claude Shannon: The Father of Information Theory



Reprinted with corrections from *The Bell System Technical Journal*,
Vol. 27, pp. 379–423, 623–656, July, October, 1948.

A Mathematical Theory of Communication

By C. E. SHANNON

INTRODUCTION

THE recent development of various methods of modulation such as PCM and PPM which exchange bandwidth for signal-to-noise ratio has intensified the interest in a general theory of communication. A basis for such a theory is contained in the important papers of Nyquist¹ and Hartley² on this subject. In the present paper we will extend the theory to include a number of new factors, in particular the effect of noise in the channel, and the savings possible due to the statistical structure of the original message and due to the nature of the final destination of the information.

Figure: Claude Elwood Shannon (4/30/1916 – 2/24/2001) was an American mathematician, electrical engineer, computer scientist and cryptographer known as the “**father of information theory**”. His landmark paper has more than **117,000** citations by Jan 2026.

Why Shannon's Paper Was Revolutionary

- Before Shannon, there was no clear way to compare communication across different media.
- It was unclear whether reliable communication was even possible in very noisy environments.
- Shannon showed that all communication systems share a common goal: transmitting bits in the presence of noise.
- He proved that reliable communication is possible despite noise, below a fundamental limit.
- His theory did not give specific codes, but proved that such codes must exist.
- This insight launched modern communication theory and guided later code designs.

Shannon's Diagram of a Communication System

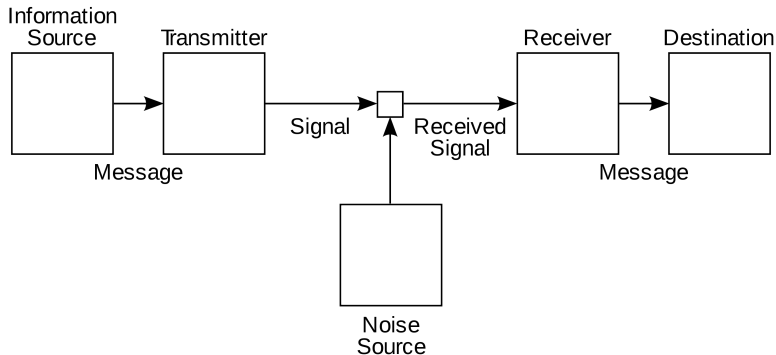


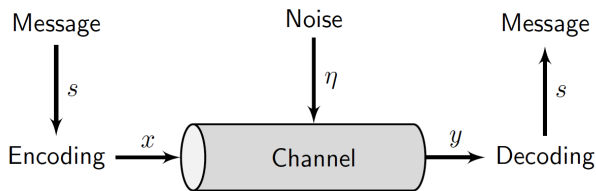
Figure: Shannon's article laid out the basic elements of communication: 1) An information source that produces a message; 2) a transmitter that operates on the message to create a signal sent via a channel; 3) a channel, which is the medium over which the signal is sent; 4) a receiver, which transforms the signal back into the message intended for delivery; and 5) a destination, for whom or which the message is intended.

Information: Physical Quantity

- Just as physics abstracts away whether energy flows through copper, air, or vacuum, Shannon abstracts away whether information flows through wires, radio waves, disks, or neurons.
- Information theory studies what can be reliably inferred from noisy observations.

A basic idea in information theory is that information can be treated very much like a physical quantity, such as mass or energy.

Claude Shannon, 1985.



Relate Information with Surprise & Probability

- Shannon defined in mathematical terms what information is.



Tomorrow: Sunny 70°F

VS



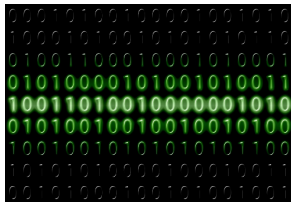
Tomorrow: Snowy 40°F



VS



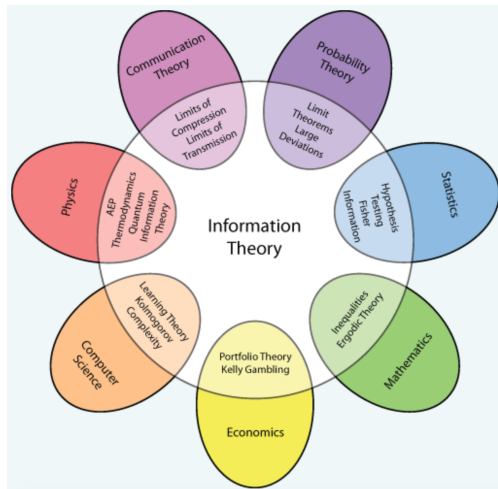
Information Bits: From Theory to Practice



- Information can be represented efficiently using **bits** (compression).
- Bits must be communicated reliably over **noisy media**.
- Shannon showed both goals are achievable and fundamentally linked.
- **Error-correcting codes** add structured redundancy to protect bits.
- This is why scratched CDs can still play music correctly.

Shannon proved the limits and optimal mechanisms for combining efficiency and reliability.

Interaction with Other Fields



- **EE (Communications).** Reliable communication is possible below channel capacity with arbitrarily small error.
- **CS (Kolmogorov Complexity).** Data complexity equals the length of its shortest generating binary program.
- **Physics (Thermodynamics).** Entropy, born in statistical mechanics, never decreases.
- **Economics (Investment).** Optimal repeated investment yields exponential wealth growth governed by entropy rate.
- **Philosophy of Science (Occam's Razor).** Simpler explanations are more likely to be correct.

Information Theory in Machine Learning and Deep Learning

- **Loss functions:** Cross-entropy and KL divergence underpin classification and probabilistic learning.
- **Representation learning:** Mutual information guides feature selection, contrastive learning, and disentanglement.
- **Regularization and generalization:** Information bottleneck, noise injection, and minimum description length (MDL) explain why models generalize.
- **Generative modeling:** Variational inference and VAEs minimize KL divergence between data and models.
- **Compression and efficiency:** Rate–distortion theory informs autoencoders, model compression, and quantization.
- **Uncertainty and exploration:** Entropy measures uncertainty in active learning and reinforcement learning.

Broad Topics (Partially Covered)

- **Information measures (entropy, divergence, mutual information):** Convexity, monotonicity and continuity properties. Extremization, saddle point, capacity as information radius.
- **Lossless data compression:** Variable length and fixed length. Linear compression. Slepian–Wolf bound. Ergodic sources: Shannon-McMillan and ergodic theorems. Universal data compression. Lempel–Ziv–Welch (LZW) algorithm.
- **Lossy data compression:** Scalar quantization and Panter-Dite approximation. Vector quantization and rate-distortion theorem. Separation principle.
- **Channel coding:** Achievability and converse bounds. Asymptotics: Capacity, strong converse, error-exponents, channel dispersion. Gaussian channels (parallel, with intersymbol interference, minimal energy-per-bit). Coding with feedback: Zero and non-zero error capacities.
- **Hypothesis testing:** Bounds for finite number of samples. Asymptotics: Stein and Chernoff exponents. Large deviations: Sanov, I-projection, tilting.

Access to Lecture Slides

- Lecture slides from the Fall 2023 offering of this course are available at [this GitHub repo](#).
- This year's lecture slides will be released progressively via [this Dropbox folder](#). Students are advised to bookmark the link and download the most recent version after each lecture.
- Most core content remains unchanged, though some slides will be updated.